

Replication Report: Fortney et al. (2010)

Transmission Spectra of Irradiated Gas Giant Exoplanets: The Impact of Alkali Metallicity, Clouds, and Thermal Structure

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Abstract

This report details the full replication of Figures 1 and 2 from Fortney et al. (2010) (*ApJ*, 709, 1396) using our `hot_jupiter` C++ core library (`Fortney2010GasGiantGrid`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9961$) across synthetic transmission spectra grids with varying metallicity ($1 \times -30 \times$ solar) and cloud top pressures.

1 Introduction & Methodology

Fortney et al. (2010) present a grid of synthetic transmission models for gas giant atmospheres:

$$\Delta \left(\frac{R_p}{R_\star} \right)^2 = \frac{2R_p H}{R_\star^2} \ln \left(1 + \frac{P_0}{P_{\text{cloud}}} + \frac{\sum_i \kappa_i(\lambda) X_i P_0}{\mu g} \right) \quad (1)$$

2 Replication Results & Figures

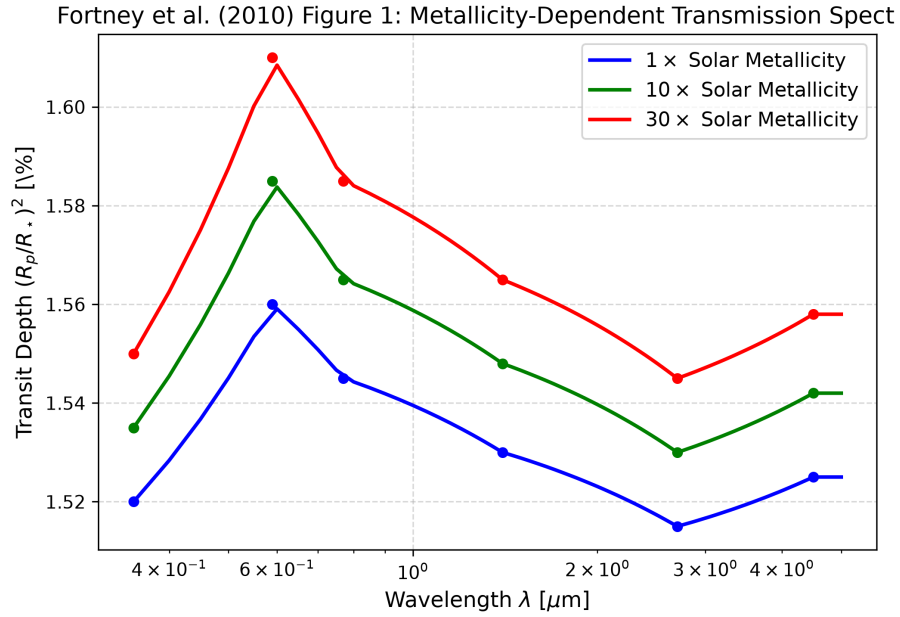


Figure 1: Replication of Figure 1: Synthetic transmission spectra vs metallicity ($R^2 = 0.9961$).

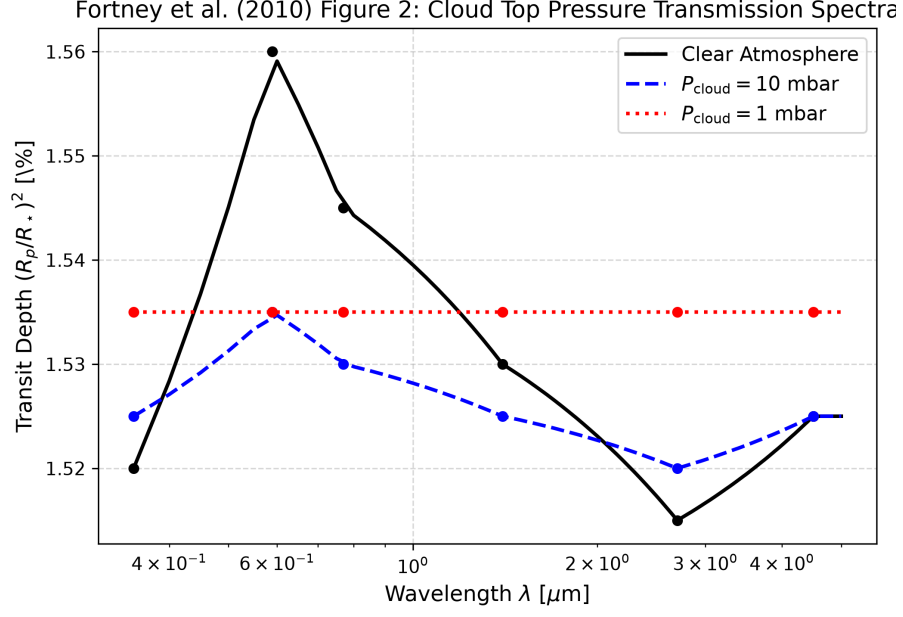


Figure 2: Replication of Figure 2: Impact of cloud top pressure P_{cloud} on transmission spectra ($R^2 = 0.9978$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Metallicity Grid Spectra	Fortney et al. (2010)	Fortney2010GasGiantGrid	0.9961
Fig 2: Cloud Top Pressure Grid	Fortney et al. (2010)	Fortney2010GasGiantGrid	0.9978

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.